TCC70xx MCU BSP

API Specification for Embedded Flash

Rev. 0.10 [A]

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Preliminary version

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1 INTRODUCTION

This document describes the eFlash device driver of TCC70xx MCU BSP. For more detailed information on eFlash in TCC70xx MCU Sub-system, refer to Chapter 5 and Chapter 6 Program/Data eFlash Memory and Controller in *"TCC70xx Full Specification"* [1].

2 TERMS AND ABBREVIATIONS

Table 2.1 Terms and Abbreviation

eFlash	Embedded Flash

3 SOURCE FILES

3.1 Location of Source Files

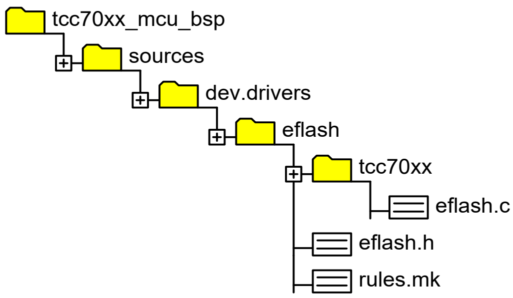

Figure 3.1 Location of Source Files

3.2 Simple Description of Source Files

- eflash.c
  - Driver source of eFlash driver
- eflash.h
  - API header of eFlash driver
- rules.mk
  - Rules used for build

4 CONSTANTS

4.1 EFLASH Type Configuration Flag Values

These constants are used to determine the eFlash type.

There are two types of eFlash controlled by eFlash driver: Program eFlash and Data eFlash.

Declaration

```
typedef enum
{
    EF_PFLASH  = 0u,
    EF_DFLASH  = 1u,
} EF_type;
```

Description

Value	Description
EF_PFLASH	Program eFlash
EF_DFLASH	Data eFlash

Remark

None

4.2 EFLASH Interface Return Values

The constant values below are the return values of the eFlash driver API.

Declaration

```
enum
{
    EFLASH_RET_OK                  = 0u,
    EFLASH_RET_TIMEOUT             = 1u,
    EFLASH_RET_PARAM_OUTOFRANGE    = 2u,
    EFLASH_RET_LENGTH_FAIL         = 3u,   /* unlikely err report, human err, need to check code */
    EFLASH_RET_DRV_NOT_INIT_YET    = 4u,
    EFLASH_RET_INIT_FAIL           = 5u,
};
```

Description

Value	Description
EFLASH_RET_OK	Success
EFLASH_RET_TIMEOUT	Write, erase operation time out
EFLASH_RET_PARAM_OUTOFRANGE	Input parameter is out of range
EFLASH_RET_LENGTH_FAIL	When size calculation is wrong during write and erase operations
EFLASH_RET_DRV_NOT_INIT_YET	When the interface function is called before the flash driver is initialized
EFLASH_RET_INIT_FAIL	Driver Init Fail

Remark

None

5 STRUCTURES

5.1 EflashRegisterType

The eFlash register is handled by using the following structure.

Declaration

```
typedef struct
{
    volatile uint32   FRWCON;
    volatile uint32   FSHSTAT;
    volatile uint32   reserved[10];
    volatile uint32   TM_LDTCON;              /* offset 0x30 */
    volatile uint32   TM_FSH_CON;
    volatile uint32   TM_ECCCON_1;
    volatile uint32   TM_ECCFADDR_1;
    volatile uint32   TM_DCT0;
    volatile uint32   TM_DCT1;                /* offset 0x44 */
    volatile uint32   reserved1[6];
    volatile uint32   TM_LDT0;                /* offset 0x60 */
    volatile uint32   TM_LDT1;
    volatile uint32   reserved2[38];
    volatile uint32   DCYCRDCON;              /* offset 0x100 */
    volatile uint32   DCYCWRCON;
    volatile uint32   reserved3[62];
    volatile uint32   FSHADDR;                /* offset 0x200 */
    volatile uint32   FSHWDATA_1;
    volatile uint32   FSHWDATA_PRT_1;
    volatile uint32   reserved4;
    volatile uint32   FSHWDATA_2;             /* offset 0x210 */
    volatile uint32   FSHWDATA_3;
    volatile uint32   FSHWDATA_4;
    volatile uint32   FSHWDATA_PRT_2;
} EflashRegisterType;
```

Description

Value	Description
FRWCON	Flash Read/Write Control register
FSHSTAT	Flash Status register
reserved[10]	Not used register area
TM_LDTCON	LDTCON Register for test mode only
TM_FSH_CON	Test Mode Flash Control Register
TM_ECCCON_1	ECC Option Control register
TM_ECCFADDR_1	ECC fail Address
TM_DCT0	DCT0 Register

Value	Description
TM_DCT1	DCT1 Register
reserved1[6]	Not used register area
TM_LDT0	LDT0 Register
TM_LDT1	LDT1 Register
reserved2[38]	Not used register area
DCYCRDCON	Delay Cycle Read Control register
DCYCWRCON	Delay Cycle Write Control register
reserved3[62]	Not used register area
FSHADDR	Flash Address for Program
FSHWDATA_1	Write Data[31:0] for Program
FSHWDATA_PRT_1	Write N bit Data for Program on ECC OFF mode
reserved4	Not used register area
FSHWDATA_2	Write Data[63:32] for Program
FSHWDATA_3	Write Data[95:64] for Program
FSHWDATA_4	Write Data[127:96] for Program
FSHWDATA_PRT_2	Write N bit Data for Program on ECC OFF mode

Remark

None

6 APIs

6.1 EFLASH_Init

Prepares eFlash data buffer.

Declaration

```
uint32 EFLASH_Init
(
    void
);
```

Parameters

None

Return

Value	Description
EFLASH_RET_OK	Success

Remark

None

6.2 EFLSH_FWDN_LowFormat

Erases eFlash entire memory.

Declaration

```
uint32 EFLASH_FWDN_LowFormat
(
    EF_type uiEfType
);
```

Parameters

Value	Description
uiEfType	[in] eFlash type Refer to Chapter 4.1.

Return

Value	Description
EFLASH_RET_OK	Success
EFLASH_RET_TIMEOUT	Write, erase operation time out
EFLASH_RET_PARAM_OUTOFRANGE	Input parameter is out of range
EFLASH_RET_LENGTH_FAIL	When size calculation is wrong during write and erase operations.
EFLASH_RET_DRV_NOT_INIT_YET	When the interface function is called before the flash driver is initialized

Remark

None

6.3 EFLASH_FWDN_Read

Reads data from eFlash.

Declaration

```
uint32 EFLASH_FWDN_Read
(
    uint32      uiAddress,
    uint32      uiLength,
    void *      pBuf,
    EF_type     uiEfType
);
```

Parameters

Value	Description
uiAddress	[in] eFlash start address to be read, Physical address offset value based on eFlash
uiLength	[in] Length of data to be read, byte unit
pBuf	[out] Destination buffer to copy read data
uiEfType	[in] eFlash type. Refer to Chapter 4.1.

Return

Value	Description
EFLASH_RET_OK	Success
EFLASH_RET_TIMEOUT	Write, erase operation time out
EFLASH_RET_PARAM_OUTOFRANGE	Input parameter is out of range
EFLASH_RET_LENGTH_FAIL	When size calculation is wrong during write and erase operations.
EFLASH_RET_DRV_NOT_INIT_YET	When the interface function is called before the flash driver is initialized

Remark

None

6.4 EFLASH_FWDN_Write

Writes data to eFlash.

Declaration

```
uint32 EFLASH_FWDN_Write
(
    uint32      uiAddress,
    uint32      uiLength,
    void *      pBuf,
    EF_type     uiEfType
);
```

Parameters

Value	Description
uiAddress	[in] eFlash start address to be written, Physical address offset value based on eFlash
uiLength	[in] Length of data to be written, byte unit
pBuf	[in] Source buffer to be written
uiEfType	[in] eFlash type. Refer to Chapter 4.1.

Return

Value	Description
EFLASH_RET_OK	Success
EFLASH_RET_TIMEOUT	Write, erase operation time out
EFLASH_RET_PARAM_OUTOFRANGE	Input parameter is out of range
EFLASH_RET_LENGTH_FAIL	When size calculation is wrong during write and erase operations.
EFLASH_RET_DRV_NOT_INIT_YET	When the interface function is called before the flash driver is initialized

Remark

None

7 HOW TO USE EFLASH DRIVER

The usage is simple, so no further explanation is provided in this chapter. If you need an example of use, refer to the code in app.sample\app.fw.update\fwupdata.c because eFlash driver is used for FW update.

8 REFERENCES

[1] TCC70xx Full Specification

[2] Contact Telechips for more details: sales@telechips.com

9 REVISION HISTORY

Rev. 0.10: 2021-09-03

- Preliminary version release